THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 09

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 9	$9^3 = 729$
2	**(b)** 400	$20^2 = 400$
3	**(d)** 4.5×10^3	One non-zero digit before the decimal point
4	**(d)** 20	Vigesimal
5	**(a)** 10 cm	$\sqrt{l^2 + b^2}$
6	**(a)** A and C	Diagonals bisect vertex angles
7	**(b)** 4	Rule for 4
8	**(b)** $a^2 - 2ab + b^2$	Standard identity
9	**(a)** 24	Product of extremes = product of means: $7x = 168$
10	**(d)** 1 : 8	15 : 120
11	**(d)** goods and services	Goods and Services Tax
12	**(b)** 36%	Multiply by 100
13	**(d)** $5\sqrt{2}$ cm	$d/\sqrt{2}$
14	**(a)** 20 cm	$\sqrt{12^2 + 16^2} = 20$
15	**(b)** ₹ 300	One part = ₹ 100
16	**(d)** the middle third of each side	Fractal construction
17	**(c)** 11.6667	Sum 70 ÷ 6
18	**(d)** 19	Top = a + 2b + c
19	**(b)** 120 cm²	$\frac{1}{2}d_1d_2$
20	**(d)** 12 cm	Area ÷ length

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Add exponents: $a^m \times a^n = a^{m+n}$ (1); e.g. $5^2 \times 5^3 = 5^5$ (1).	1+1
22	$19 = 10011_2$ (2).	2
23	Last three digits $392 = 8 \times 49$, so yes (2).	2
24	Inverse relation: $8 \times 15/12 = 10$ days (2).	2
25	If $a^2 + b^2 = c^2$ in a triangle then it is right-angled with hypotenuse c (2).	2
26	60, 120, 180 (2).	2
27	Ordered: 1, 3, 5, 7, 9, 12, 15; median 7 (2).	2
28	77 cm² (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	LCM $= 60 = 2^2 \times 3 \times 5$ (1); need $2^2 \times 3^2 \times 5^2$ (1); square $= 900$ (1).	1+1+1
30	Only one pair of opposite sides is parallel in a trapezium (3).	3
31	$x^2 + 12x + 35$ (2); at $x = 1$: $(6)(8) = 48 = 1 + 12 + 35$ (1).	2+1
32	Salary ₹ 30000 (2); annual savings ₹ 54000 (1).	2+1
33	12 edge cubes (3).	3
34	$2n + (2n + 2) + (2n + 4) = 6n + 6 = 6(n + 1)$ (3).	3
35	By the theorem, height $= \sqrt{10^2 - 6^2}$ (2); $= 8$ m (1).	2+1
	OR: Diagonal 29 m (1.5); saved $41 - 29 = 12$ m (1.5).	1½+1½

36	Trapezium $= \frac{1}{2}(14+22)\times 7 = 126$ cm² (2); rectangle 154 cm² (1).	2+1
	OR: Area 126 m² (1.5); perimeter 46 m, cost ₹ 1380 (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $10^{-5}/10^{-10} = 10^{5}$ times (3). (ii) 7×10^{-3} mm (2).	3+2
	OR: (i) Compact, easy to compare and calculate (2). (ii) 9.1×10^{-31} kg (3).	2+3
38	(i) 37.5% and 62.5% (2). (ii) 28 L (3).	2+3
	OR: (i) $1.1\times 1.1 = 1.21$ (3). (ii) $1.1\times 0.9 = 0.99$, a 1% decrease (2).	3+2
39	(i) Mean 38.75; median $(38+45)/2 = 41.5$ (2). (ii) 90 (1). (iii) Large variation (2).	2+1+2
	OR: (i) Both are multiplied by 2 (3). (ii) Mean 5 to 10, median 5 to 10 (2).	3+2
40	(i) $40+18 = 58$ units² (3). (ii) Subtract a missing rectangle from a larger one (2).	3+2
	OR: (i) Four congruent right triangles (3). (ii) 16 cm (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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